

Sep 21, 2026

Data Visualization

Week 1. Introduction

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Why “Data Visualization” ?

These factors necessitate that Data Visualization be considered as a separate course:

- The increasing volume of data due to declining storage costs in parallel with technological advancements, and the need to quickly derive insights from this data,
- The effectiveness of visual representation compared to other explanatory techniques,
- The growing demand in the industry for skilled professionals who can extract meaning from data using visual tools.

Aim

- Understanding data visualization methods.
- Practicing data storytelling techniques.
- Gaining awareness of bad visualization habits.

Self-Thinking

How many ways can we use to present data?

Why “Visual Representation” ?

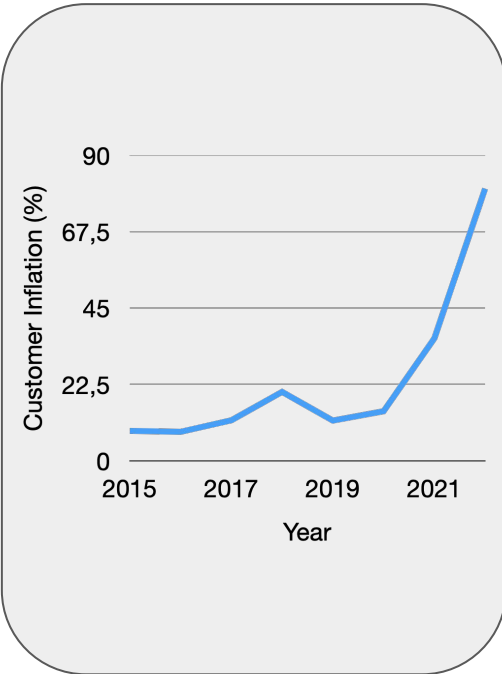
Text

Since 2015, consumer inflation in Turkey has increased each year compared to the previous year. In 2015, inflation rose by 8.81% compared to the previous year, followed by increases of 8.53%, 11.92%, 20.3%, 11.84%, 14.6%, 36.08%, and 80.21% in the subsequent years.

Table

Year	Consumer Inflation (%)
2015	8.91
2016	8.53
2017	11.92
2018	20.3
2019	11.84
2020	14.6
2021	36.08
2022	80.21

Plot



Why “Visual Representation” ?

The human brain processes visual data 60,000 times faster than text data.*

*<https://oit.williams.edu/files/2010/02/using-images-effectively.pdf>

Applications

Throughout the course, the exercises will be conducted using R.

posit Cloud (<https://posit.cloud>) will be used for this purpose.

You are free to use any other tool!

Course Materials

All materials can be accessed through the course's GitHub page:

https://github.com/mcavs/Course_DataVisualization

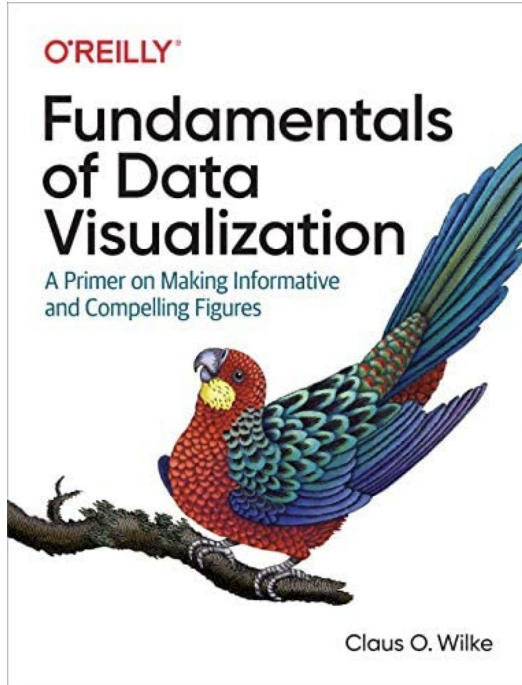
Grading

- 10 Hands-on exercises - 3 of them: *cornerstones of final project* (50%)
 - project draft #1 (title + abstract + data source + RQs) - minor changes can be done
 - project draft #2 (some plots)
 - project draft #3 (more plots + draft report)
- Project - *poster presentation in the last week* (50%)
 - an open event for 2 hours
 - a poster summarize your project in the size of A1
 - GitHub repository (report + code + data + poster)

Timeline

Week 1 Sep 21 Hands-on Exercise 1 Practical Lab	Week 2 Sep 28 Hands-on Exercise 1 Practical Lab	Week 3 Oct 05 Hands-on Exercise 2 Practical Lab	Week 4 Oct 12 Hands-on Exercise 3 Practical Lab	Week 5 Oct 19 Project Milestone #1 Key Deliverable	Week 6 Oct 26 Hands-on Exercise 4 Practical Lab	Exam Nov 02 1st Midterm No Class
Week 7 Nov 09 Hands-on Exercise 5 Practical Lab	Week 8 Nov 16 Hands-on Exercise 6 Practical Lab	Week 9 Nov 23 Project Milestone #2 Key Deliverable	Week 10 Nov 30 Hands-on Exercise 7 Practical Lab	Exam Dec 07 2nd Midterm No Class	Week 11 Dec 14 Project Milestone #3 Final Pre-Review	Final Dec 21 Project Poster Presentations Poster Showcase

Book



It is open source and free:
<https://clauswilke.com/dataviz/>

Practice

Eskişehir Monthly Maximum Temperatures (2017 – 2026)

Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
2017	11.2	14.8	21.0	25.4	29.1	34.6	38.2	36.5	32.1	24.3	17.5	13.8
2018	13.5	16.2	22.4	27.8	28.6	32.9	36.8	36.2	31.5	25.0	18.2	11.4
2019	10.8	13.9	20.1	23.6	30.5	33.8	36.1	37.4	32.0	27.2	21.1	14.2
2020	12.0	15.6	21.8	24.5	33.2	32.5	37.0	36.9	35.8	28.4	18.0	15.1
2021	14.1	17.0	20.5	26.2	31.8	34.2	38.6	37.8	30.4	24.1	19.3	14.5
2022	9.5	14.2	18.6	26.0	29.4	33.5	37.5	38.1	32.6	26.3	18.7	15.6
2023	15.2	13.8	22.7	25.1	28.7	33.0	39.4	40.2	33.8	27.9	20.4	14.8
2024	14.6	17.5	23.1	28.4	30.2	36.1	39.8	39.5	33.1	26.8	17.9	13.2
2025	13.8	16.0	21.5	27.1	31.4	35.2	39.1	38.8	32.5	26.0	19.0	12.6
2026	13.2	16.8	22.0	26.5	31.0	35.8	39.5	39.2	—	—	—	—

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Flexible Homeworks

1. Please learn and practice how to pull request in GitHub
2. Create a posit free account